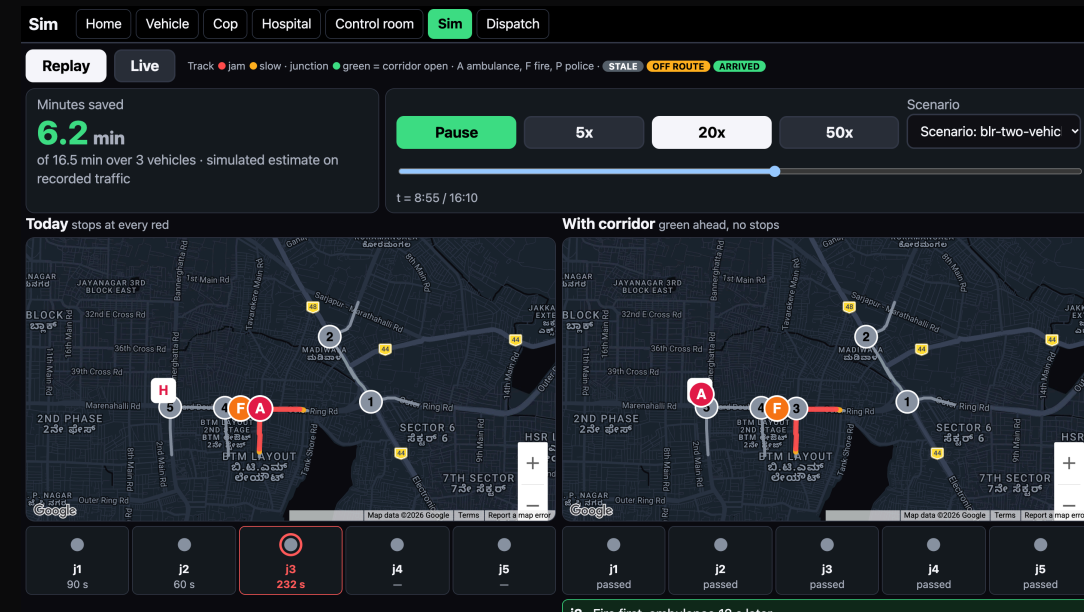


# Emergency Green Corridor

Every red light costs a life.

Tell the cop at the next junction, early enough, that an ambulance or fire engine is coming.

Built by Nandish · Live at [green-corridor-2026.web.app](https://green-corridor-2026.web.app)



## The problem, seen daily on Hosur Road

---

**7–10%**

survival lost per minute in cardiac arrest without defibrillation

**2–3 km**

covered in 30–50 min by Bengaluru ambulances at peak hours

**60%+**

of Karnataka 108 cardiac, stroke and respiratory calls missed the 10-min target

The officer at the junction has no idea they are coming. Full signal preemption needs city controller integration: years away.

Sources: American Heart Association, CPR Facts and Stats ([cpr.heart.org](https://www.cpr.heart.org)) and AHA, Circulation, 2017 · The Siasat Daily, 28 Jun 2026 (Karnataka Health Minister announcement) · CAG audit of Karnataka 108 EMS, 2014–19, reported by The News Minute, Dec 2020 (state-wide, not Bengaluru only)

# The 20-second demo

**Today:** stops at every red.

**With corridor:** cop warned early, queue cleared.

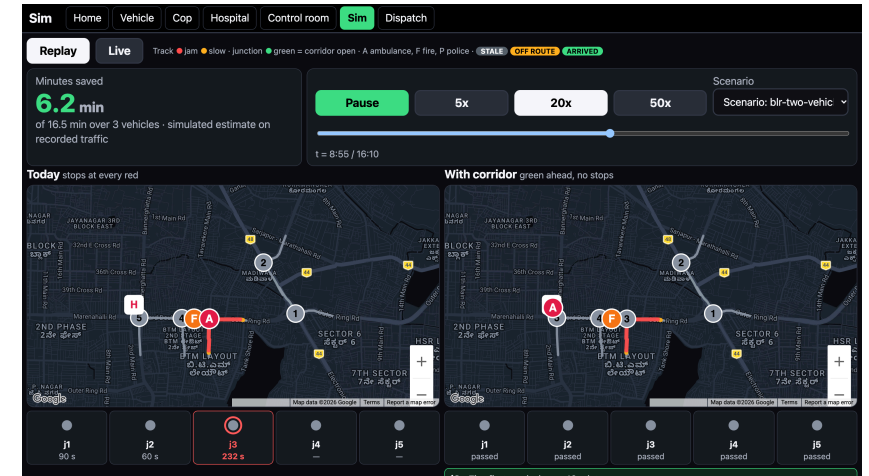
Silk Board → Jayadeva Hospital, the same scripted scenario in both lanes: GPS ticks generated along the real corridor roads, with hand-authored traffic spans (a 500 m queue at junction 3, 100 m at junction 4). The 3 vehicles run as one scenario: a critical ambulance, an ambulance following it as a platoon, and a fire engine.

≈ **16.5 min**

saved across 3 vehicles

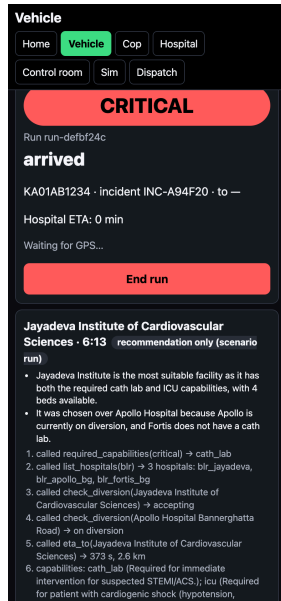
**3**

critical ambulance, platoon  
ambulance, fire engine

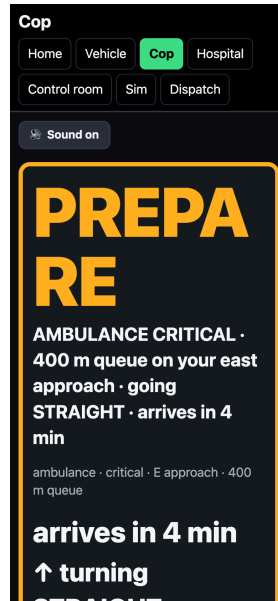


Simulated baseline on a scripted scenario, not a field measurement: per junction passed, cycle/4 + queue/2 m/s. Figure from the scenario replay ( `blr-two-vehicles` ). Live runs use live Routes traffic; scenario runs use the scripted spans.

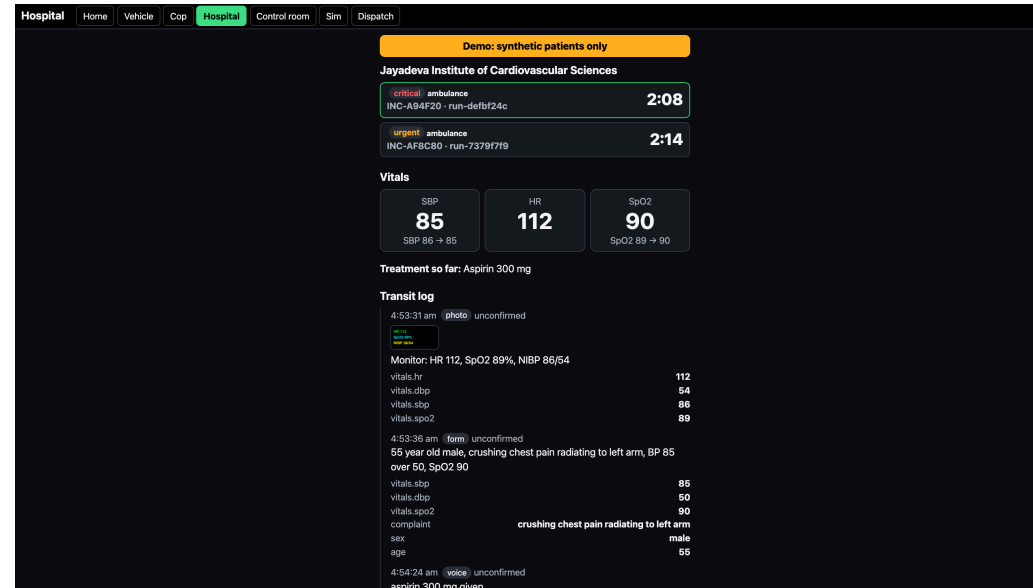
# What it does: six screens



/vehicle  
voice, tier tap



/cop spoken  
alert, one ACK



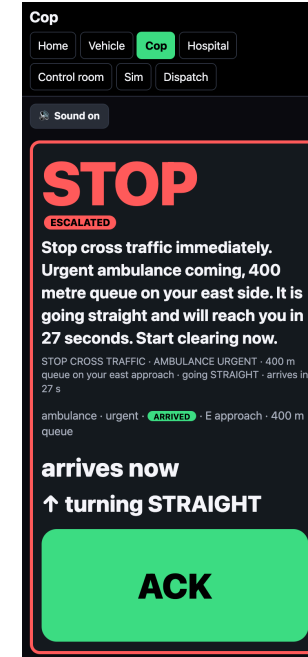
/hospital live log, vitals, treatment so far (the ATMIST brief lands at ETA-5)

Also: /dispatch incident IDs · /control all runs and escalations · /sim with-vs-without replay.

# How a cop gets warned: the lead-time engine

Live runs: Routes traffic spans (scenario runs: scripted spans) → queue metres → clearance time → two-stage alert.

```
jam_m    = Σ JAM m + 0.5 × Σ SLOW m
clear_s  = 20 + jam_m / 2.0
eta_s    = 0.5 × routes_eta
          + 0.5 × distance / speed_60s
PREPARE          eta_s <= clear_s + 15
STOP CROSS TRAFFIC eta_s <= 30
```



No ACK in 20 s raises an escalation flag and an audit entry.

# Rules decide, Gemini explains

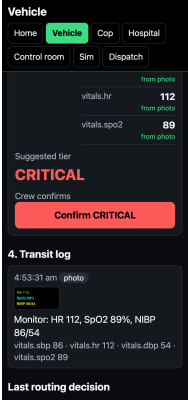
---

| Step          | Who                      | What                                                                                                                     |
|---------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1.<br>Extract | Gemini                   | speech or monitor photo → fixed JSON fields, never a score                                                               |
| 2. Tier       | <code>acuity.py</code>   | deterministic lookup → critical / urgent / stable; crew confirms with one tap                                            |
| 3. Order      | <code>priority.py</code> | sort by (tier, ETA): a <b>sequence, not a hold</b>                                                                       |
| 4.<br>Explain | Gemini                   | rewords the rule-built sentence for the cop ("Fire engine first, ambulance 12 s later"); validated, template if it fails |

A language model must not decide who lives. Rules are auditable, testable and carry self-checks.

# Where Gemini is load-bearing

| Job                             | What                                                                                                              |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Voice and photo extraction      | speech, typed text or a monitor photo into a fixed schema, with no tools.                                         |
| ATMIST handover                 | the hospital brief and prep checklist, generated at ETA minus 5 minutes.                                          |
| Alert phrasing                  | the short spoken line each junction cop hears.                                                                    |
| Sequencing sentence             | Gemini rewords a rule-built template into one line; the result is validated and the template is used if it fails. |
| Cop voice notes to rule actions | a spoken report from the junction fills a fixed schema, and plain rules act on it.                                |
| After-action report             | a plain summary of each finished run, with the timeline built in code.                                            |
| Hospital routing agent          | built on Agent Development Kit, with four tools and a code guard that checks its choice before it is applied.     |



87%  
field accuracy

73%  
tier accuracy

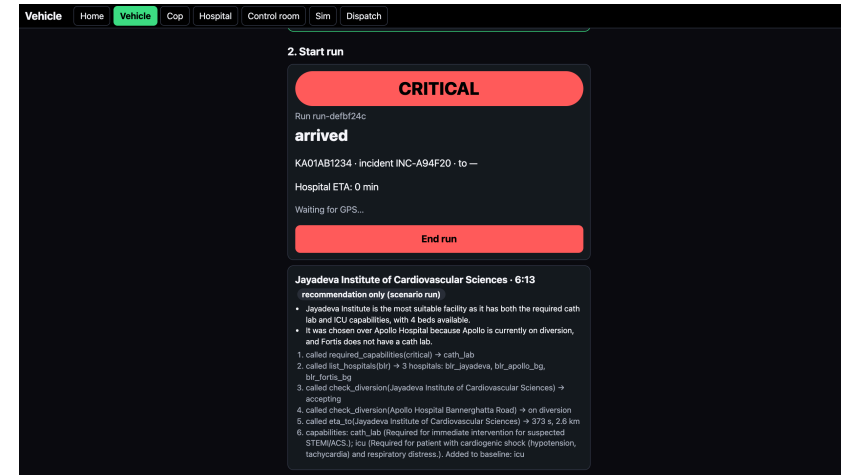
Accuracy: synthetic voices (Cloud Text-to-Speech), 12 clips (11 scored), 6 Oct 2026; not field recordings, replaced by real EMT clips when recorded.  
Extraction: `gemini-3.1-flash-lite` , fallback `gemini-3-flash-preview` . Synthetic patients only; a clinician confirms every extracted value.

# The agent: hospital routing on Agent Development Kit

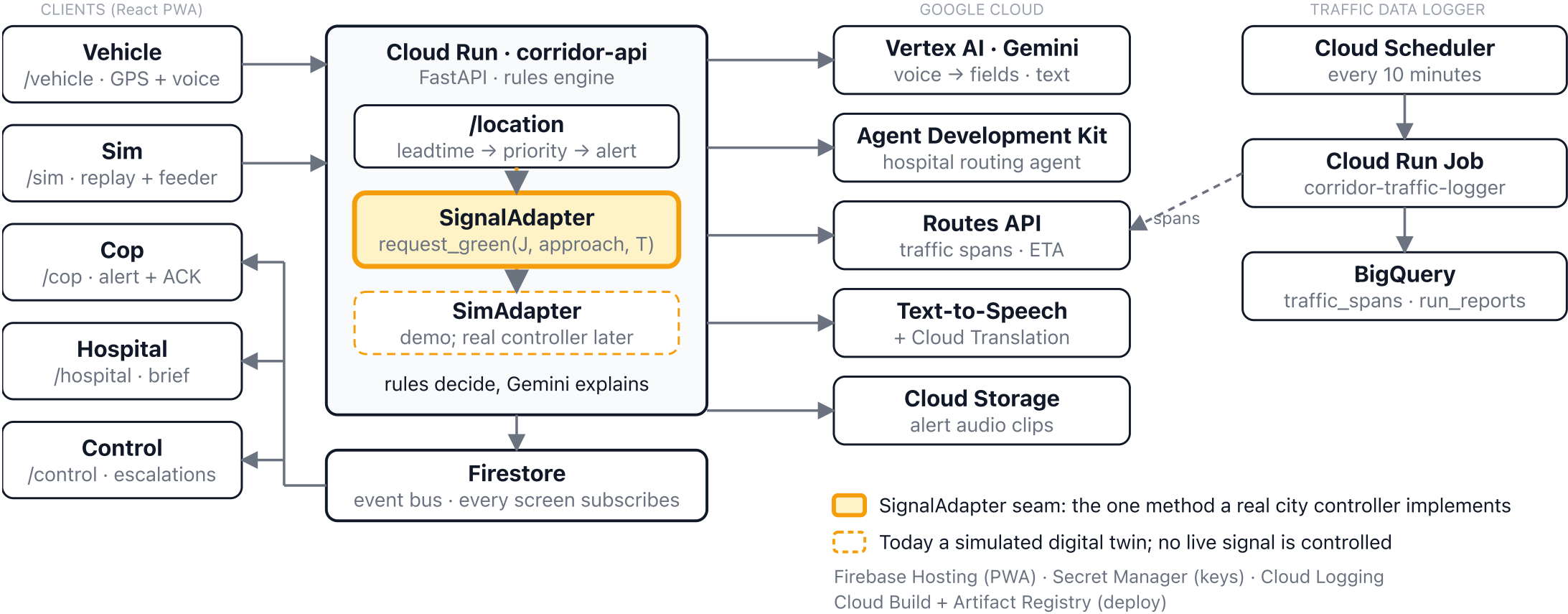
```
required_capabilities(critical) -> cath_lab  
list_hospitals(blur) -> 3 hospitals  
check_diversion(Jayadeva) -> accepting  
check_diversion(Apollo) -> on diversion  
eta_to(Jayadeva) -> 373 s, 2.6 km  
Added to baseline: icu
```

Destination Jayadeva, confidence 1.0. Alternatives:  
Apollo (diversion), Fortis (no cath lab).

An ADK agent on Vertex AI picks the destination after the crew confirms the tier, with up to two rejected alternatives and a confidence. `required_capabilities` is a keyword baseline for validation. A code guard re-checks the choice; any failure or 20 s timeout falls back to the nearest eligible hospital.



# Architecture



Cost by design: ~3 traffic-aware Routes calls per vehicle-minute; AI side effects run off the request path. Wired on main only. BigQuery ML: pipeline in place, retrained before submission on N rows ([fill at freeze]); not a deployed forecast.

# Real data: the traffic logger

---

## BUILT

- Cloud Scheduler → Cloud Run Job during Bengaluru peak hours (08–11, 17–21 IST), running since 6 Oct
- Live jam and slow spans per approach → BigQuery `traffic_spans`, two corridors
- Same `jam_metres` as the live engine

## PROOF OF PIPELINE, NOT A FORECAST

- Rows so far: N ([fill at freeze]). The first 216 rows were midnight with zero queues
- BigQuery ML model: a pipeline proof until peak rows accumulate; retrained before submission, not a deployed forecast
- First step to a learning controller; clearance rate stays constant until observed

Run reports (the with-vs-without report card per run) also land in BigQuery `run_reports`.

# Sustainability

---

**0.76 L/h**

idle burn, small petrol car (low end of 0.2–0.5 gal/h)

**2.35 kg**

CO<sub>2</sub> per litre of petrol (diesel 2.69 kg/L)

**0.6 kg**

CO<sub>2</sub> idled by one 20-vehicle queue in 60 s (illustration)

Preemption also idles the cross traffic, so the claim is the **net** effect. We size the green to the queue, so cross-traffic idling is bounded by `clear_s`.

`net_fuel_L = (ambulance-lane idling avoided - cross-traffic idling added) × idle_L_per_s`

**SDG 3.8** emergency care access (universal health coverage) · **SDG 11.2** sustainable transport.

Idle burn: Argonne National Laboratory (Gaines, Rask, Keller), US DOE, 2014. CO<sub>2</sub>: US EPA, GHG Equivalencies Calculator, 8,887 g/gal petrol and 10,180 g/gal diesel, 2024. The per-queue arithmetic (20 vehicles, 60 s) is an illustration, not a measurement or a net saving.

# Deployment path, and honesty

---

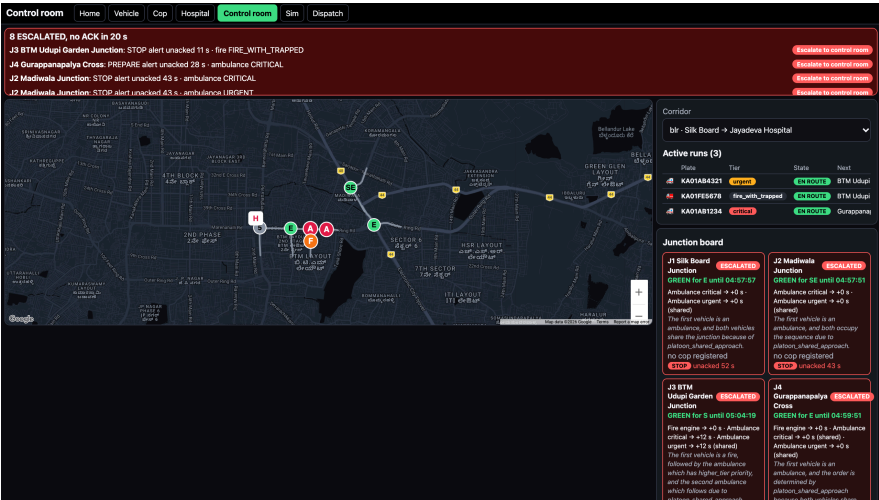
| Phase          | What                                              |
|----------------|---------------------------------------------------|
| <b>Today</b>   | cop alerts, no hardware, a phone                  |
| <b>Phase 2</b> | real controller behind <code>SignalAdapter</code> |
| <b>Phase 3</b> | learning controller on junction data              |

## STATED PLAINLY

- Traffic signals are simulated behind an adapter; the demo scenario uses hand-authored traffic spans
- Patients are synthetic; no user accounts; device-scoped tokens protect vehicle, cop and run actions; demo endpoints are rate-limited but reachable
- Minutes saved is a simulation estimate

# Global by configuration

|                | blr                   | hyd             |
|----------------|-----------------------|-----------------|
| Corridor       | Silk Board → Jayadeva | Hyderabad       |
| Alert language | English, Kannada      | English, Telugu |
| Switch         | ?corridor=blr         | ?corridor=hyd   |



A corridor is one JSON file: junctions, approaches, signal cycle, language. No code change.

# Try it

---

## Live

[green-corridor-2026.web.app](https://green-corridor-2026.web.app)

## Repo

[github.com/Nandish3010/ideal-disco](https://github.com/Nandish3010/ideal-disco)

API health: [corridor-api-919512130399.asia-south1.run.app/health](https://corridor-api-919512130399.asia-south1.run.app/health) · Demo video: [\[link, 3 minutes\]](#)

**Ask:** feedback from traffic police and 108 operators, and a pilot junction.